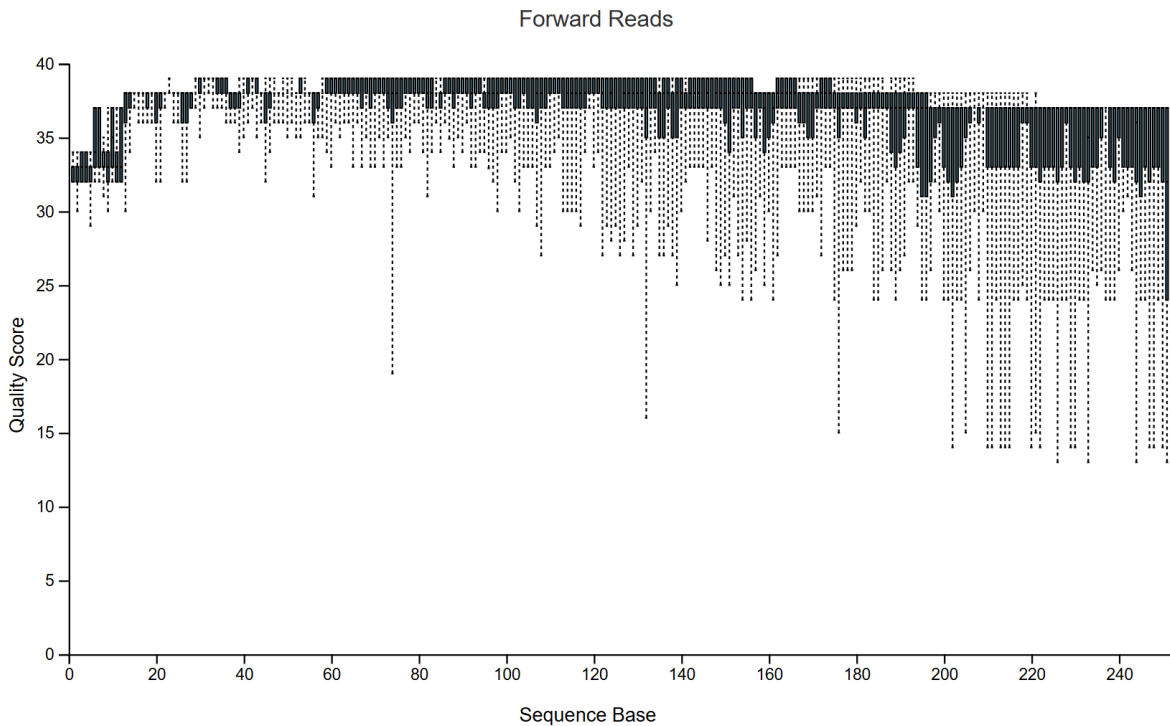


Microbiome Section Report

1. Looking at the metadata file, I think that the three columns that would be useful for later would be the population, sex, and flock columns because they give categorical data which is what the alpha and beta diversity metrics uses.
- 2.



For p-trunc-len I chose a value of 220 and for p-trim-left I chose a value of 5 because I did not want to lose too many reads but I also didn't want to keep the low quality reads over 220/

- 3.

Sequence Length Statistics

Download sequence-length statistics as a TSV

Sequence Count	Min Length	Max Length	Mean Length	Range	Standard Deviation
1881	215	345	232.3	130	8.73

view File: <code>table.qzv</code> × ≡	
Retained 383,746 (100.00%) features in 24 (100.00%) samples at the specified sampling depth.	
Sample ID	Frequency
188_S347_L001	48,410
136_S299_L001	44,257
162_S315_L001	37,633
301_S383_L001	32,985
308_S339_L001	31,091
277_S363_L001	25,458
370_S355_L001	15,834
233_S379_L001	15,697
229_S328_L001	14,979
59_S304_L001	13,978
109_S323_L001	12,924
209_S308_L001	12,840
183_S293_L001	12,114
47_S354_L001	11,374
320_S307_L001	11,035
351_S306_L001	10,070
230_S317_L001	9,279
137_S352_L001	9,118

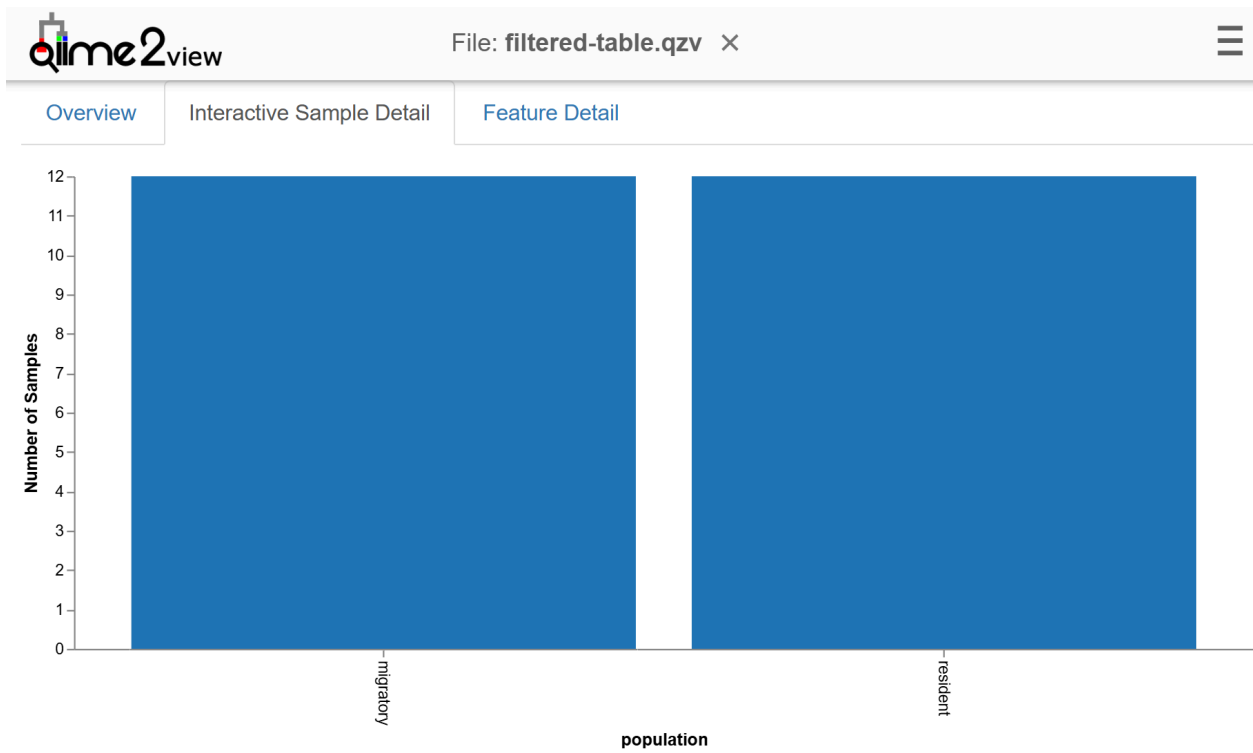
137_S352_L001	9,118
116_S378_L001	4,524
81_S376_L001	3,984
155_S341_L001	2,336
89_S331_L001	1,716
256_S377_L001	1,302
299_S330_L001	808

Yes there are about eight samples that are below 10,000, those sample reads have frequency lengths from 808-9,279.

4. If we sort by confidence the top hits are: featureID 07f2bc59c802429da211a0129b298046 taxon: k_Bacteria;p_Actinobacteria;c_Thermoleophilia;o_Solirubrobacterales;f_Patulibacteraceae;g_;s_. If we sort by taxon the top hit is feature id:b94c5b424c5ac13e0ba4f1eacc0b3368 and taxon k_Archae;p_Crenarchaepta;c_Thaumarchaepta;o_Nitrososphaerales;f_Nitrososphaeraceae;g_Candidatus Nitrososphaera;s_gargensis.

5. Level 3 taxonomy refers to class, so everything on the graph is sorted out by class. The diversity slightly changes as we move by category but the Gammaproteobacteria remains the majority throughout.

6.



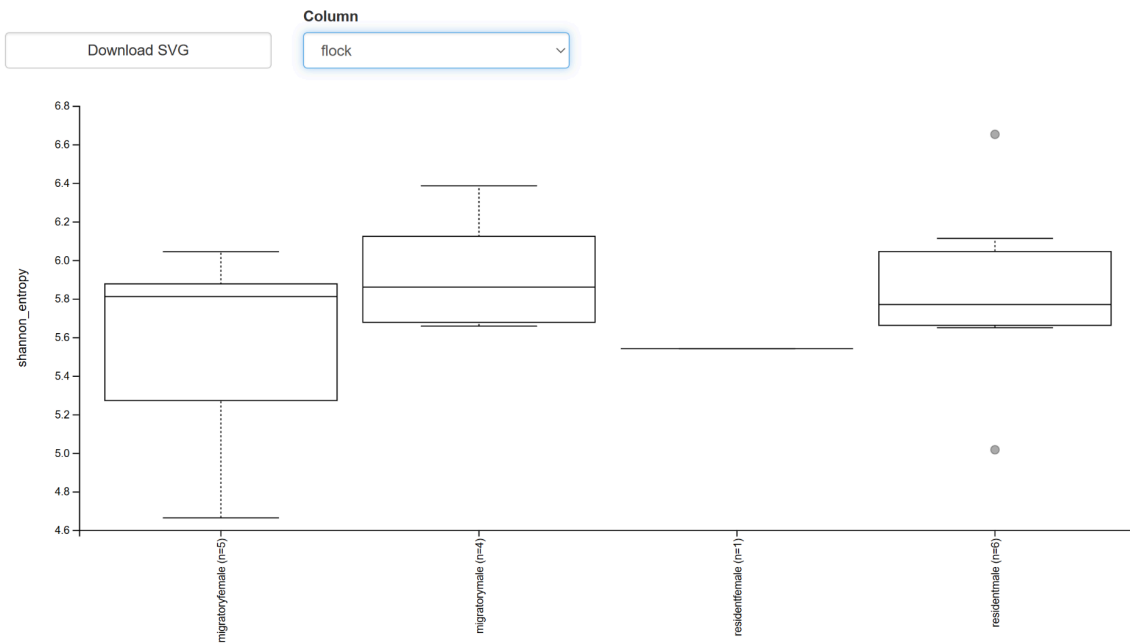
 view File: filtered-table.qzv ×	
188_S347_L001	45,927
136_S299_L001	42,710
162_S315_L001	36,199
308_S339_L001	27,876
301_S383_L001	26,578
277_S363_L001	22,792
229_S328_L001	13,362
59_S304_L001	13,055
370_S355_L001	12,773
233_S379_L001	12,660
209_S308_L001	11,641
109_S323_L001	11,242
47_S354_L001	10,623
183_S293_L001	9,563
230_S317_L001	8,327
320_S307_L001	8,185
351_S306_L001	7,282
137_S352_L001	7,147
81_S376_L001	3,599
116_S378_L001	3,232
155_S341_L001	1,977
89_S331_L001	1,258
256_S377_L001	1,048
299_S330_L001	623

In order to keep most of my samples and still maintain the high quality reads I've chosen to set my cutoff value to 8000, this will keep all of my more reliable samples and it won't drop my diversity results down.

7. Alpha diversity is how much diversity there is within a single sample and how they are distributed. Observed features shows how many different microbes there are and Shannon diversity shows how many types there are and how spread out they are as well.

8. shannon:

Alpha Diversity Boxplots



Observed:



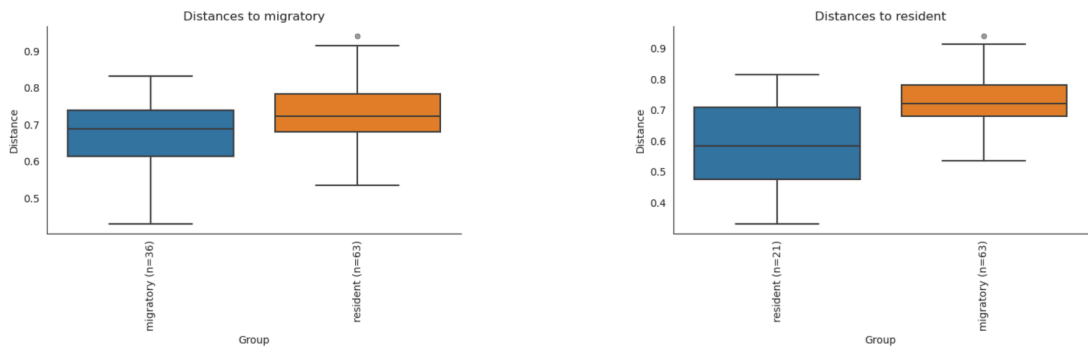
Since all of the q values were <0.05 there weren't any statistically significant differences suggesting that the diversity is consistent.

9.

qime2view	File: bray-curtis-population-significance.qzv	Visualization	Citations	Provenance	Metadata
number of permutations		999			

Group significance plots

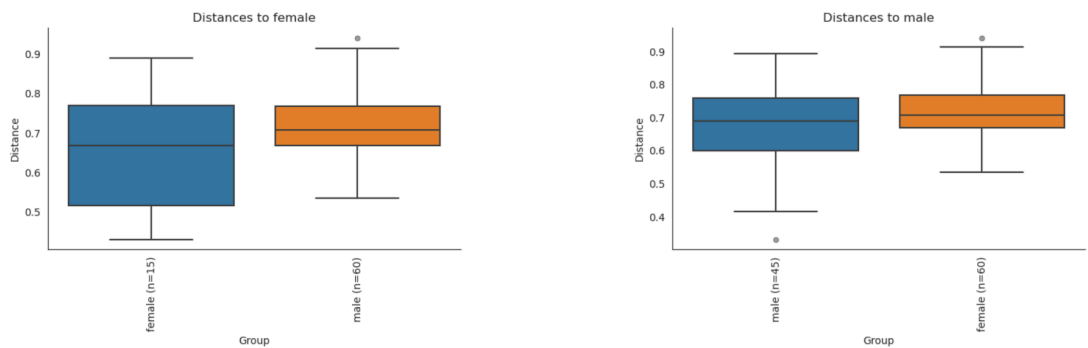
[Download raw data as TSV](#)



qime2view	File: bray-curtis-sex-significance.qzv	Visualization	Citations	Provenance	Metadata
number of groups		2			
test statistic		2.119424			
p-value		0.019			
number of permutations		999			

Group significance plots

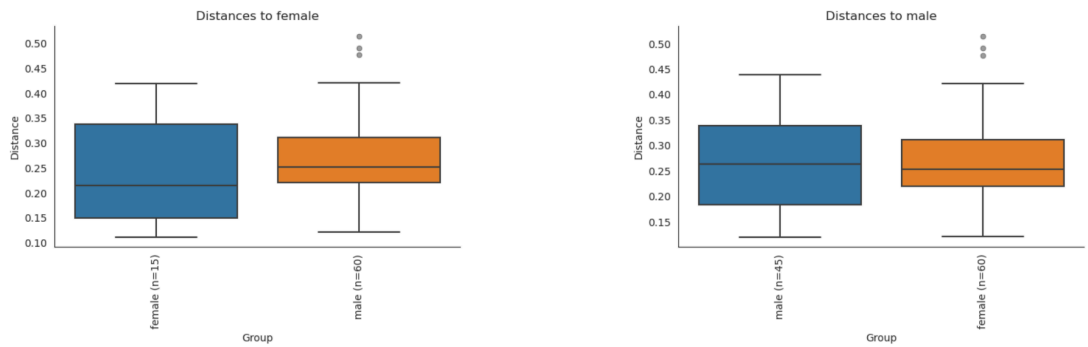
[Download raw data as TSV](#)



qime2view	File: weighted-unifrac-sex-significance.qzv	Visualization	Citations	Provenance	Metadata
number of groups		2			
test statistic		1.617464			
p-value		0.18			
number of permutations		999			

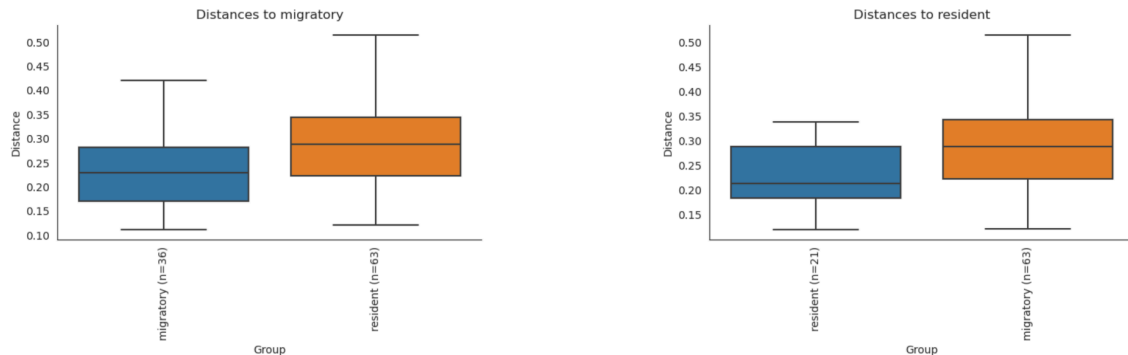
Group significance plots

[Download raw data as TSV](#)



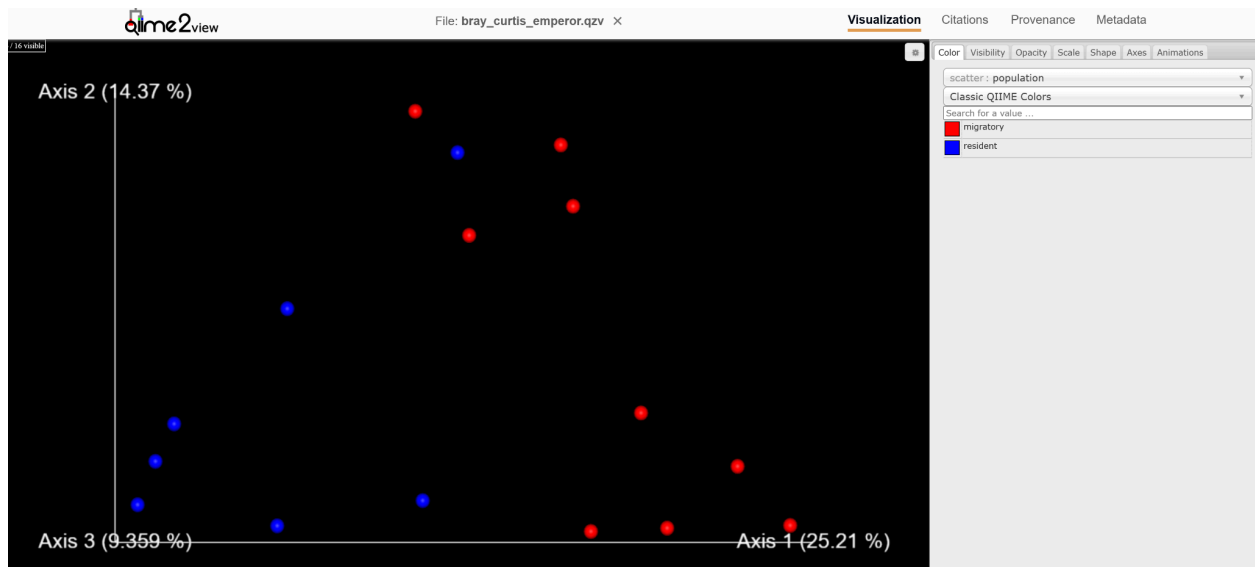
Group significance plots

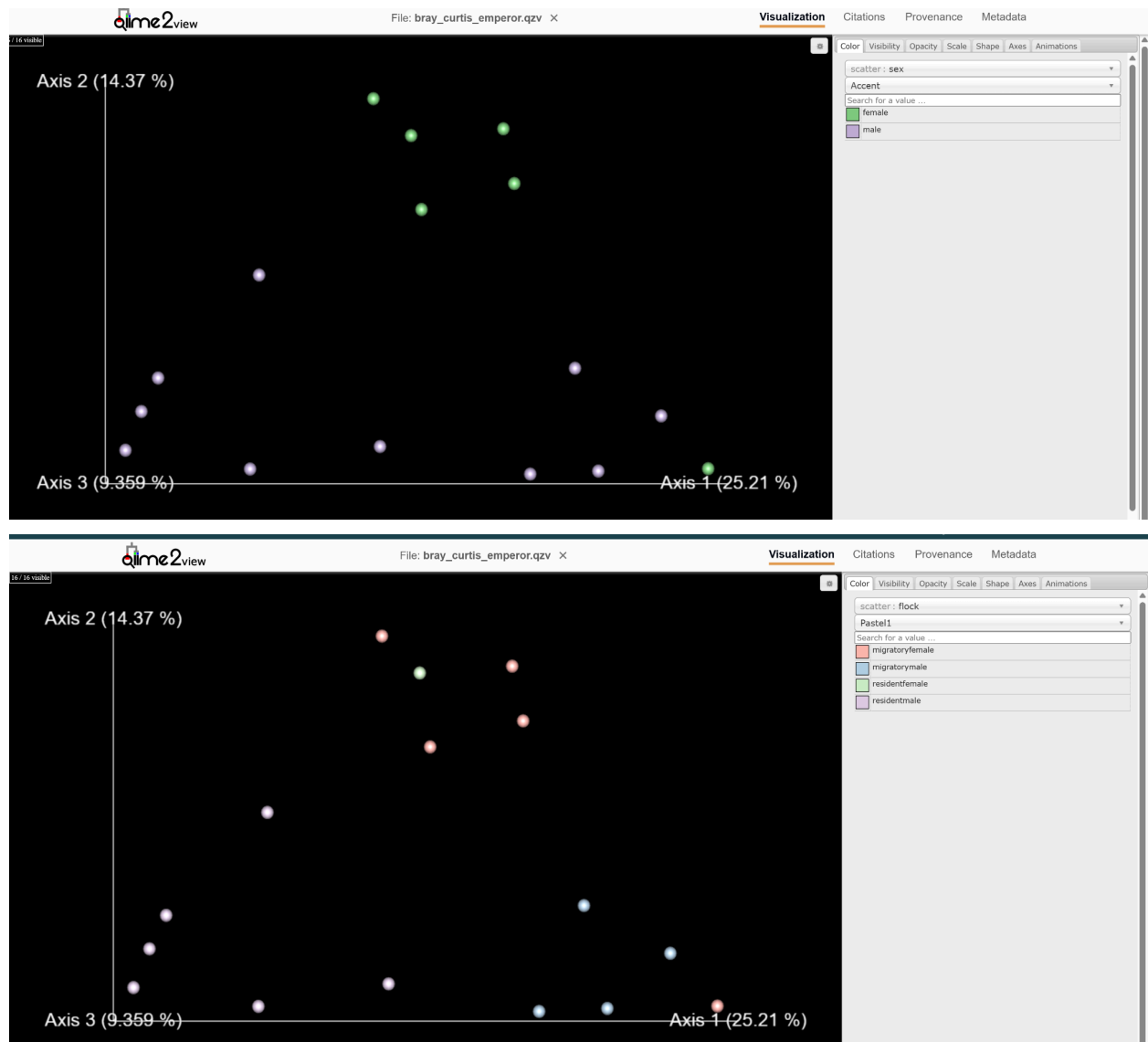
[Download raw data as TSV](#)



Beta diversity looks at the differences **between** samples which helps understand whether or not the microbiomes are the same. I chose to compare populations and sex using Bray-Curtis and Weighted Unifrac. Bray-Curtis found significant differences between males and females meaning that the types and amounts of microbes varied but there weren't significant differences between migratory or resident birds. Weighted unifrac didn't find any significant differences between sex and population. Weighted unifrac looks at how related the microbes are, so even though the amounts between can vary, the kinds of microbes present are still similar.

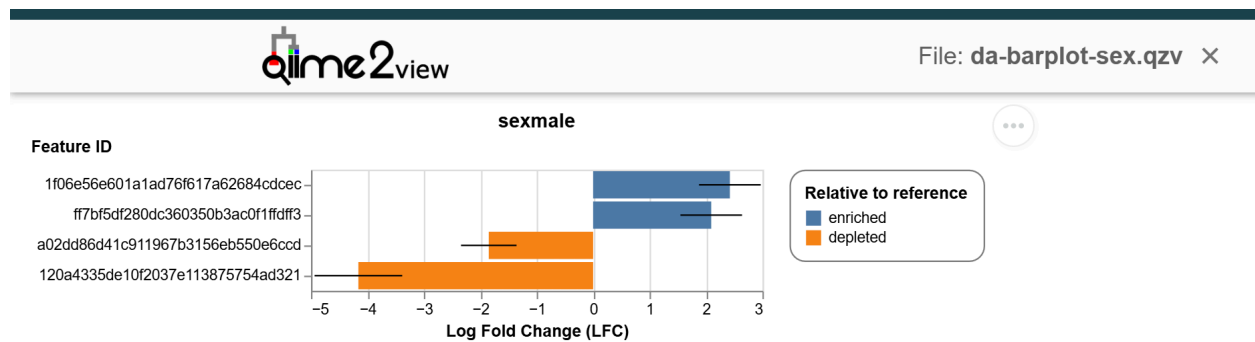
10.





These images help make sense of the question before because there's a visual aid to help us see that there's a clear separation between male and females whereas in the other two graphs the dots are a lot closer to each other.

11.



There are 4 differentially expressed taxa in males. There are 2 enriched taxa in males and 2 depleted taxa in males (more abundant in females). The two enriched taxa are k__Bacteria; p__Bacteroidetes; c__Sphingobacteriia; o__Sphingobacteriales; f__Sphingobacteriaceae; g__; s__ and k__Bacteria; p__Bacteroidetes; c__Sphingobacteriia; o__Sphingobacteriales; f__NS11-12; g__; s__. The two depleted taxa are k__Bacteria; p__Proteobacteria; c__Betaproteobacteria; o__Burkholderiales; f__Comamonadaceae and k__Bacteria; p__Proteobacteria; c__Gammaproteobacteria